

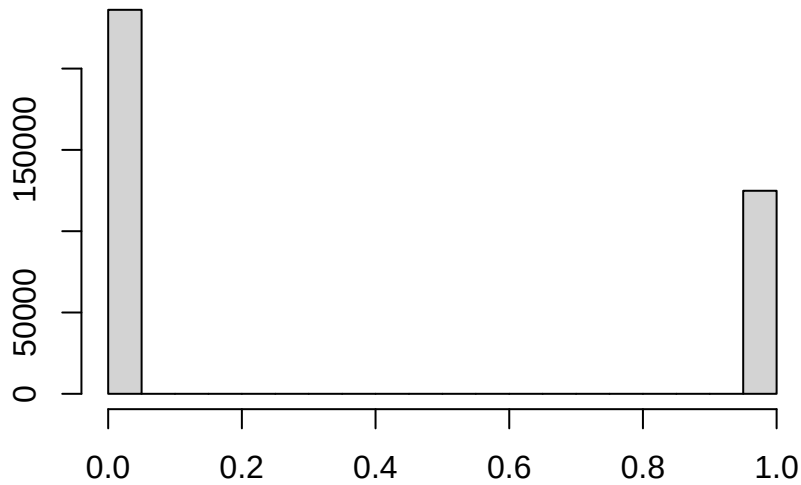
Competition – Contractor Experience

2024-06-05

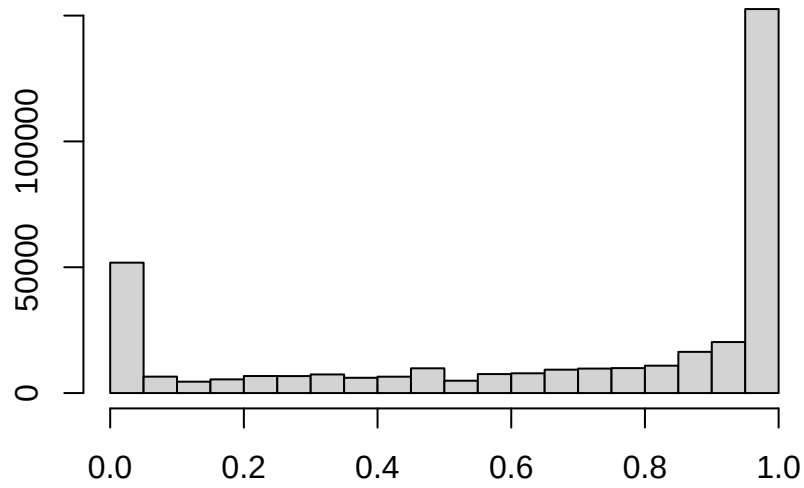
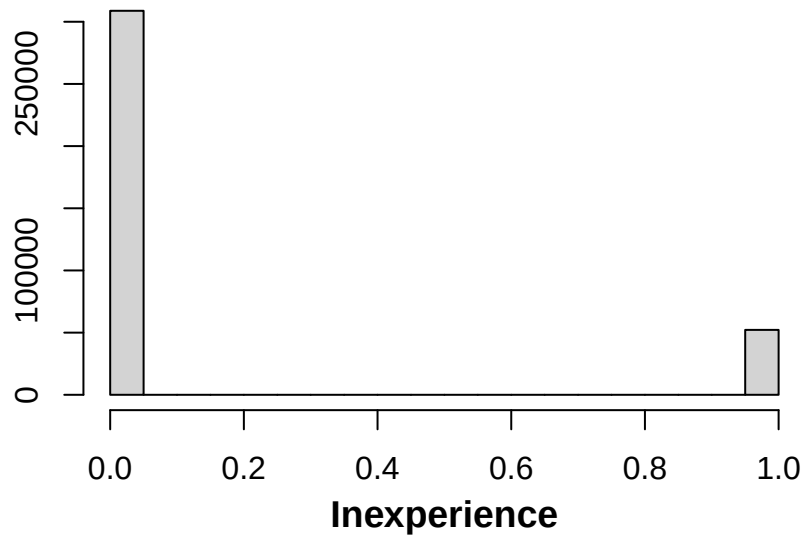
1 Time Invariant Status

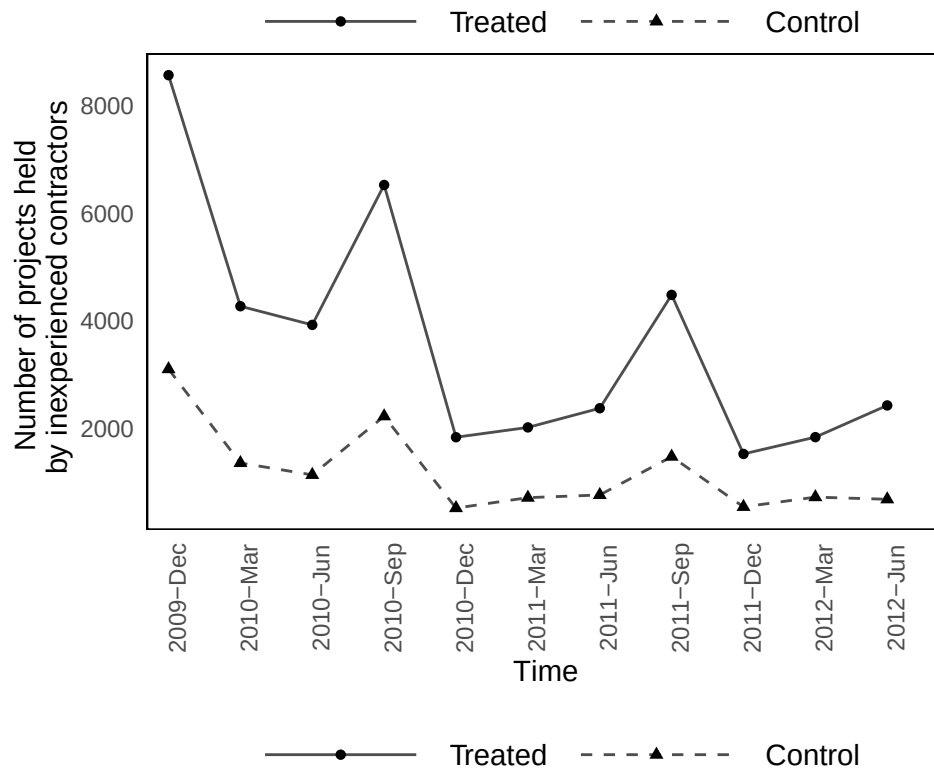
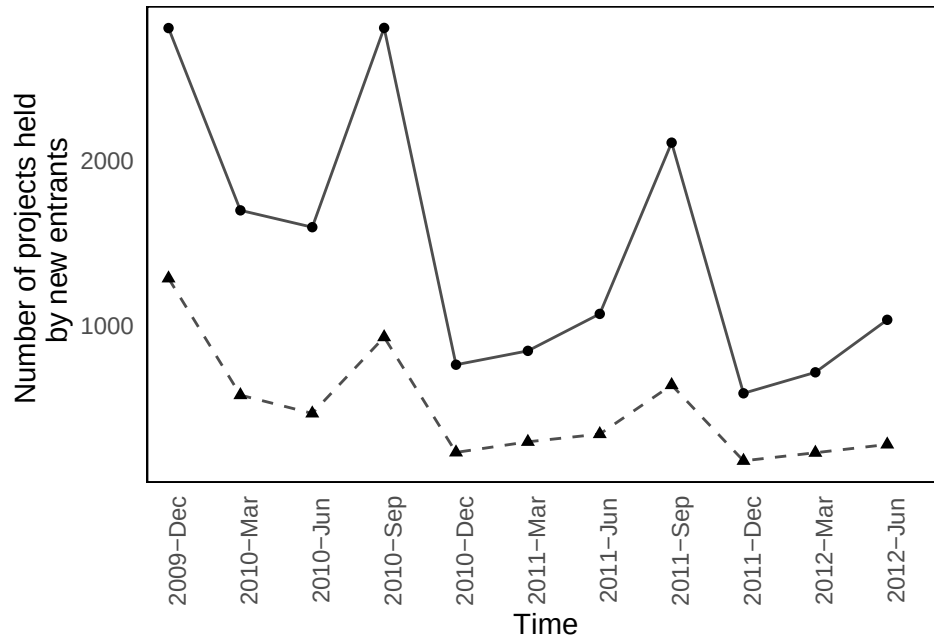
- Let n_i^{task} = Number of projects of the same task held by the contractor in the three quarters before starting project i
- Let n_i = Number of projects held by the contractor in the three quarters before starting project i
- Define:
 - Inexperienced contractor $_i$ = 1 if $n_i^{task} = 0$
 - New entrant $_i$ = 1 if $n_i = 0$
 - Level of inexperience $_i$ = $1 - \frac{n_i^{task}}{n_i}$ if $n_i > 0$
 - Level of inexperience $_i$ = 1 if $n_{i,k,t} = 0$.

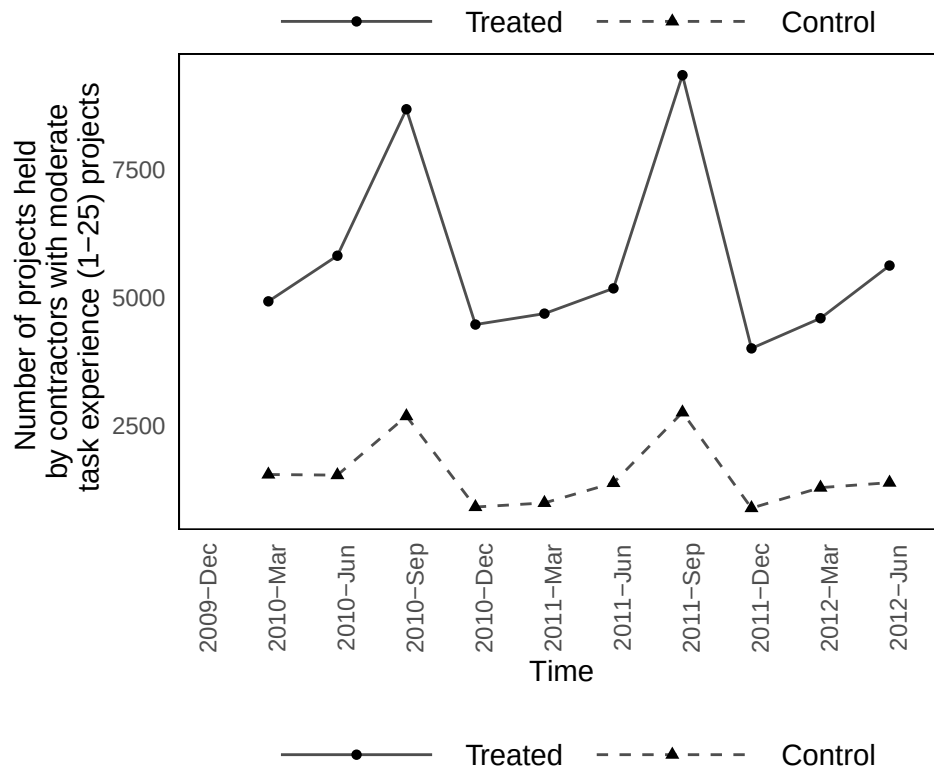
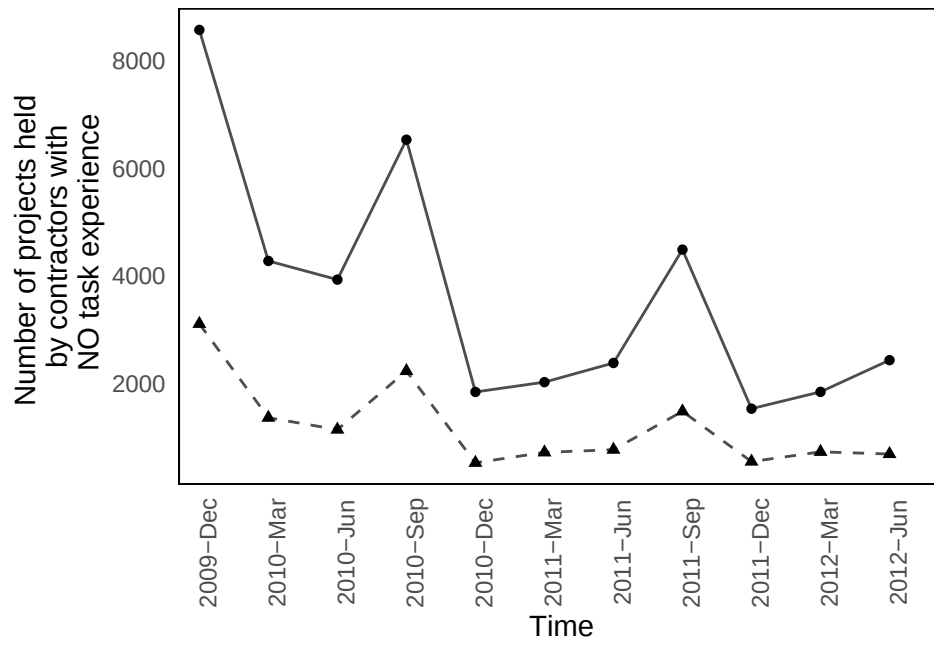
Task inexperience



New Entrant







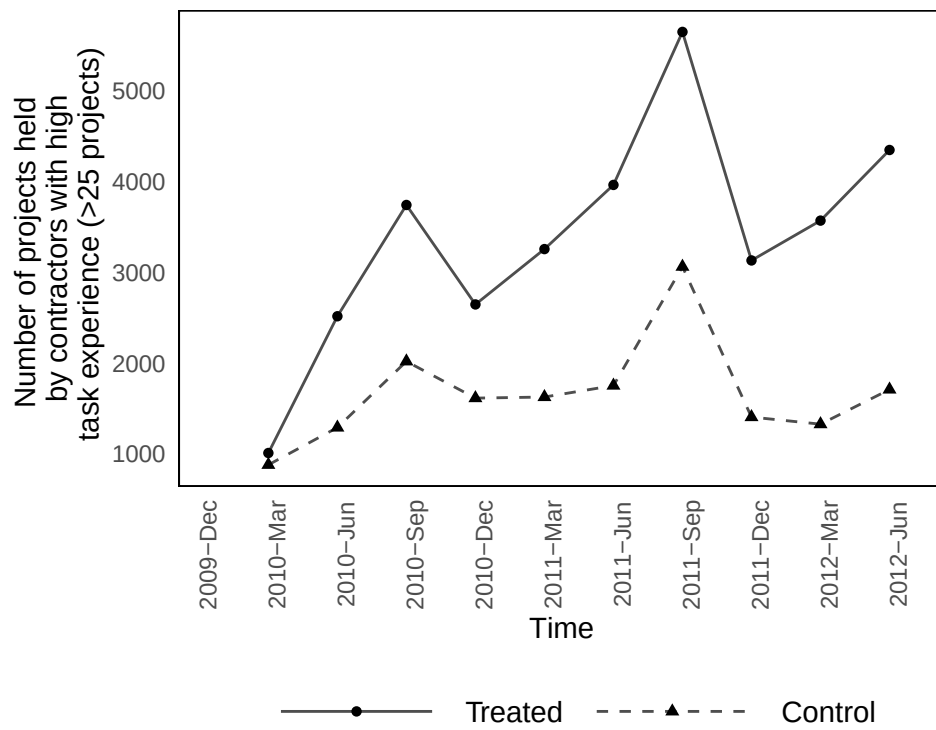


Table 1: Logit model

	Inexperienced contractor						
	I	II	III	IV	V	VI	VII
Constant	-0.272*** (0.015)	0.181*** (0.023)					
Treat	0.028 (0.018)	0.066*** (0.024)	0.088*** (0.025)	0.262*** (0.034)	0.225*** (0.035)	0.187*** (0.036)	0.201*** (0.037)
Post	-0.694*** (0.018)	-0.771*** (0.022)					
Treat $\times$ Post	-0.075*** (0.021)	-0.064** (0.026)	-0.081*** (0.026)	0.048 (0.031)	0.069** (0.031)	0.078** (0.034)	0.055 (0.036)
Ln(Stage)		0.140*** (0.005)	0.180*** (0.005)	0.241*** (0.006)	0.243*** (0.006)	0.288*** (0.007)	0.293*** (0.007)
Num offers						0.011*** (0.003)	0.004 (0.004)
Ln(Initial Budget)						0.042*** (0.008)	0.040*** (0.009)
Ln(Initial Duration)						0.354*** (0.022)	0.505*** (0.026)
Post $\times$ Num offers							0.013*** (0.004)
Post $\times$ Ln(Initial Budget)							0.004 (0.008)
Post $\times$ Ln(Initial Duration)							-0.260*** (0.026)
Time FE	No	No	Yes	Yes	Yes	Yes	Yes
Task FE	No	No	No	Yes	Yes	Yes	Yes
Industry FE	No	No	No	No	Yes	Yes	Yes
Observations	361,013	184,976	178,669	175,095	175,078	156,327	156,327
Squared Correlation	0.03	0.04	0.04	0.25	0.27	0.30	0.30
Pseudo R ²	0.02	0.03	0.03	0.21	0.22	0.25	0.25
BIC	454,413.69	240,067.46	228,585.31	187,362.22	185,863.22	157,302.90	157,141.70

Clustered (Project ID) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

2 Regression results

	Percentage delay rate						
	I	II	III	IV	V	VI	VII
Constant	5.381*** (0.151)	9.934*** (0.161)					
Treat	-2.638*** (0.170)	-2.423*** (0.164)	-2.441*** (0.164)	-0.367** (0.163)	-0.419*** (0.162)	-0.497*** (0.151)	-0.497*** (0.155)
Inexperienced contractor	0.203 (0.221)	-0.204 (0.210)	-0.121 (0.211)	-0.411** (0.200)	-0.418** (0.200)	1.278*** (0.183)	1.414*** (0.184)
Post	0.021 (0.179)	-0.955*** (0.173)					
Treat $\times$ Inexperienced contractor	0.748*** (0.251)	0.958*** (0.239)	0.953*** (0.239)	-0.264 (0.224)	-0.197 (0.224)	-0.552*** (0.206)	-0.585*** (0.206)
Inexperienced contractor $\times$ Post	0.844*** (0.281)	0.725*** (0.269)	0.655** (0.270)	0.045 (0.262)	0.003 (0.261)	0.094 (0.255)	-0.095 (0.257)
Treat $\times$ Post	0.820*** (0.203)	0.967*** (0.196)	0.992*** (0.196)	0.636*** (0.185)	0.636*** (0.185)	0.798*** (0.177)	0.806*** (0.188)
Treat $\times$ Inexperienced contractor $\times$ Post	-0.020 (0.324)	-0.475 (0.312)	-0.479 (0.312)	-0.242 (0.301)	-0.206 (0.301)	0.377 (0.301)	0.418 (0.301)
Ln(Stage)		2.829*** (0.030)	2.870*** (0.031)	2.770*** (0.030)	2.748*** (0.030)	2.666*** (0.030)	2.649*** (0.030)
Num offers						-0.006 (0.008)	-0.041*** (0.011)
Ln(Initial Budget)						0.429*** (0.021)	0.430*** (0.028)
Ln(Initial Duration)						-7.400*** (0.092)	-7.806*** (0.126)
Post $\times$ Num offers							0.057*** (0.015)
Post $\times$ Ln(Initial Budget)							0.000 (0.032)
Post $\times$ Ln(Initial Duration)							0.659*** (0.157)
Time FE	No	No	Yes	Yes	Yes	Yes	Yes
Task FE	No	No	No	Yes	Yes	Yes	Yes
Industry FE	No	No	No	No	Yes	Yes	Yes
Observations	184,908	184,882	184,882	184,882	184,882	165,473	165,473
R ²	0.01	0.06	0.06	0.12	0.12	0.22	0.22
Within R ²			0.06	0.05	0.05	0.15	0.15

Clustered (Project ID) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 2: Logit model

	I	II	III	New Entrant IV	V	VI	VII
Constant	-1.503*** (0.021)	-1.164*** (0.030)					
Treat	-0.076*** (0.024)	-0.053 (0.032)	-0.048 (0.033)	0.297*** (0.043)	0.304*** (0.044)	0.272*** (0.047)	0.271*** (0.047)
Post	-0.512*** (0.024)	-0.520*** (0.029)					
Treat $\times$ Post	0.104*** (0.028)	0.120*** (0.034)	0.118*** (0.034)	0.158*** (0.038)	0.174*** (0.038)	0.195*** (0.042)	0.200*** (0.043)
Ln(Stage)		0.114*** (0.007)	0.133*** (0.007)	0.149*** (0.008)	0.149*** (0.008)	0.193*** (0.009)	0.193*** (0.009)
Num offers						0.014*** (0.004)	0.002 (0.005)
Ln(Initial Budget)						0.032*** (0.010)	0.027** (0.011)
Ln(Initial Duration)						0.213*** (0.026)	0.238*** (0.030)
Post $\times$ Num offers							0.022*** (0.005)
Post $\times$ Ln(Initial Budget)							0.010 (0.009)
Post $\times$ Ln(Initial Duration)							-0.050 (0.032)
Time FE	No	No	Yes	Yes	Yes	Yes	Yes
Task FE	No	No	No	Yes	Yes	Yes	Yes
Industry FE	No	No	No	No	Yes	Yes	Yes
Observations	361,013	184,976	184,976	182,001	181,977	162,648	162,648
Squared Correlation	0.01	0.01	0.02	0.18	0.19	0.21	0.21
Pseudo R ²	0.01	0.01	0.02	0.18	0.20	0.21	0.21
BIC	296,200.67	165,117.57	163,976.91	141,579.17	140,278.91	118,834.39	118,828.23

Clustered (Project ID) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

	Percentage delay rate						
	I	II	III	IV	V	VI	VII
Constant	5.472*** (0.123)	9.929*** (0.136)					
Treat	-2.474*** (0.139)	-2.244*** (0.132)	-2.257*** (0.132)	-0.558*** (0.136)	-0.575*** (0.136)	-0.624*** (0.126)	-0.633*** (0.130)
New Entrant	0.044 (0.280)	-0.369 (0.265)	-0.336 (0.265)	-0.042 (0.256)	-0.036 (0.257)	1.222*** (0.228)	1.310*** (0.228)
Post	0.084 (0.149)	-0.880*** (0.144)					
Treat $\times$ New Entrant	1.090*** (0.324)	1.509*** (0.307)	1.483*** (0.306)	0.102 (0.294)	0.133 (0.295)	-0.710*** (0.271)	-0.736*** (0.270)
New Entrant $\times$ Post	1.254*** (0.371)	1.166*** (0.355)	1.136*** (0.355)	0.269 (0.346)	0.211 (0.346)	0.185 (0.337)	0.043 (0.339)
Treat $\times$ Post	0.799*** (0.170)	0.886*** (0.164)	0.904*** (0.164)	0.699*** (0.157)	0.690*** (0.157)	0.896*** (0.152)	0.917*** (0.163)
Treat $\times$ New Entrant $\times$ Post	-0.570 (0.431)	-1.172*** (0.415)	-1.151*** (0.414)	-0.601 (0.404)	-0.550 (0.404)	0.434 (0.409)	0.465 (0.409)
Ln(Stage)		2.840*** (0.030)	2.885*** (0.031)	2.746*** (0.030)	2.725*** (0.030)	2.694*** (0.030)	2.679*** (0.030)
Num offers						-0.006 (0.008)	-0.039*** (0.011)
Ln(Initial Budget)						0.435*** (0.021)	0.435*** (0.028)
Ln(Initial Duration)						-7.366*** (0.092)	-7.748*** (0.125)
Post $\times$ Num offers							0.054*** (0.015)
Post $\times$ Ln(Initial Budget)							0.002 (0.032)
Post $\times$ Ln(Initial Duration)							0.622*** (0.156)
Time FE	No	No	Yes	Yes	Yes	Yes	Yes
Task FE	No	No	No	Yes	Yes	Yes	Yes
Industry FE	No	No	No	No	Yes	Yes	Yes
Observations	184,908	184,882	184,882	184,882	184,882	165,473	165,473
R ²	0.01	0.06	0.06	0.12	0.12	0.22	0.22
Within R ²			0.06	0.05	0.05	0.15	0.15

Clustered (Project ID) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

3 Regressions – Continuous variable

	Level of inexperience						
	I	II	III	IV	V	VI	VII
Constant	0.720*** (0.003)	0.797*** (0.004)					
Treat	0.029*** (0.003)	0.029*** (0.004)	0.033*** (0.004)	-0.004 (0.004)	-0.007* (0.004)	-0.017*** (0.004)	-0.014*** (0.004)
Post	-0.123*** (0.003)	-0.147*** (0.004)					
Treat × Post	0.014*** (0.004)	0.015*** (0.004)	0.012*** (0.004)	0.031*** (0.004)	0.032*** (0.004)	0.032*** (0.004)	0.028*** (0.004)
Ln(Stage)		0.018*** (0.001)	0.021*** (0.001)	0.025*** (0.001)	0.025*** (0.001)	0.026*** (0.001)	0.026*** (0.001)
Num offers						0.005*** (0.000)	0.004*** (0.000)
Ln(Initial Budget)						0.001 (0.001)	0.002** (0.001)
Ln(Initial Duration)						0.044*** (0.002)	0.052*** (0.003)
Post × Num offers							0.001** (0.000)
Post × Ln(Initial Budget)							-0.002** (0.001)
Post × Ln(Initial Duration)							-0.013*** (0.003)
Time FE	No	No	Yes	Yes	Yes	Yes	Yes
Task FE	No	No	No	Yes	Yes	Yes	Yes
Industry FE	No	No	No	No	Yes	Yes	Yes
Observations	361,013	184,976	184,976	184,976	184,976	165,567	165,567
R ²	0.02	0.04	0.06	0.27	0.27	0.30	0.30
Within R ²			0.01	0.01	0.01	0.02	0.02

Clustered (Project ID) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

	Percentage delay rate						
	I	II	III	IV	V	VI	VII
Constant	6.369*** (0.286)	11.066*** (0.282)					
Treat	-2.594*** (0.333)	-2.327*** (0.317)	-2.373*** (0.317)	0.222 (0.304)	0.121 (0.303)	-0.310 (0.283)	-0.297 (0.284)
Level of inexperience	-1.156*** (0.336)	-1.525*** (0.320)	-1.564*** (0.321)	-0.765** (0.307)	-0.744** (0.306)	1.071*** (0.282)	1.264*** (0.282)
Post	-0.850*** (0.327)	-1.800*** (0.313)					
Treat $\times$ Level of inexperience	0.461 (0.389)	0.540 (0.370)	0.582 (0.370)	-0.938*** (0.349)	-0.833** (0.347)	-0.507 (0.324)	-0.550* (0.323)
Level of inexperience $\times$ Post	1.505*** (0.398)	1.409*** (0.380)	1.460*** (0.381)	0.854** (0.364)	0.773** (0.365)	0.593* (0.354)	0.327 (0.352)
Treat $\times$ Post	1.356*** (0.383)	1.582*** (0.367)	1.633*** (0.368)	0.884** (0.346)	0.896*** (0.345)	1.055*** (0.334)	1.057*** (0.337)
Treat $\times$ Level of inexperience $\times$ Post	-1.010** (0.465)	-1.361*** (0.445)	-1.404*** (0.445)	-0.629 (0.423)	-0.623 (0.422)	-0.258 (0.414)	-0.211 (0.413)
Ln(Stage)		2.868*** (0.031)	2.917*** (0.031)	2.771*** (0.030)	2.749*** (0.030)	2.687*** (0.030)	2.671*** (0.030)
Num offers						-0.008 (0.008)	-0.042*** (0.011)
Ln(Initial Budget)						0.435*** (0.021)	0.433*** (0.028)
Ln(Initial Duration)						-7.381*** (0.092)	-7.759*** (0.126)
Post $\times$ Num offers							0.056*** (0.015)
Post $\times$ Ln(Initial Budget)							0.005 (0.032)
Post $\times$ Ln(Initial Duration)							0.614*** (0.157)
Time FE	No	No	Yes	Yes	Yes	Yes	Yes
Task FE	No	No	No	Yes	Yes	Yes	Yes
Industry FE	No	No	No	No	Yes	Yes	Yes
Observations	184,908	184,882	184,882	184,882	184,882	165,473	165,473
R ²	0.00	0.06	0.06	0.12	0.12	0.22	0.22
Within R ²			0.06	0.05	0.05	0.15	0.15

Clustered (Project ID) standard-errors in parentheses

Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

4 Bidding effect

4.1 Old vs New Entrant

	Percentage delay rate							
	Old entrants				New entrants			
	I	II	III	IV	V	VI	VII	VIII
Constant	9.900*** (0.142)				10.912*** (0.296)			
Treat	-2.246*** (0.132)	-2.255*** (0.132)	-0.387*** (0.138)	-0.415*** (0.138)	-0.516* (0.273)	-0.551** (0.271)	-0.729** (0.304)	-0.781** (0.307)
SA _i	0.700*** (0.176)	1.207*** (0.190)	1.205*** (0.189)	1.155*** (0.188)	2.685*** (0.497)	3.880*** (0.519)	3.924*** (0.558)	3.863*** (0.557)
Ln(Stage)	2.822*** (0.035)	2.910*** (0.038)	2.756*** (0.036)	2.726*** (0.036)	3.790*** (0.085)	4.026*** (0.092)	3.884*** (0.095)	3.860*** (0.095)
Post	-1.232*** (0.173)				-0.940** (0.369)			
Treat × Post	0.442** (0.197)	0.449** (0.197)	0.180 (0.190)	0.171 (0.190)	-0.618 (0.420)	-0.575 (0.420)	0.118 (0.432)	0.101 (0.432)
Treat × SA _i × Post	0.725*** (0.202)	0.718*** (0.202)	0.802*** (0.199)	0.811*** (0.198)	0.237 (0.562)	0.172 (0.561)	-0.233 (0.596)	-0.205 (0.596)
Time FE	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Task FE	No	No	Yes	Yes	No	No	Yes	Yes
Industry FE	No	No	No	Yes	No	No	No	Yes
Observations	154,121	154,121	154,121	154,121	30,761	30,761	30,761	30,761
R ²	0.06	0.06	0.13	0.13	0.07	0.07	0.14	0.14
Within R ²		0.05	0.05	0.05		0.07	0.06	0.06

Clustered (Project ID) standard-errors in parentheses

Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

5 Experienced vs Inexperienced

	Percentage delay rate							
	Experienced contractors				Inexperienced contractors			
	I	II	III	IV	V	VI	VII	VIII
Constant	9.648*** (0.172)				10.713*** (0.193)			
Treat	-2.437*** (0.164)	-2.428*** (0.164)	-0.037 (0.169)	-0.097 (0.168)	-1.365*** (0.173)	-1.397*** (0.172)	-0.903*** (0.187)	-0.891*** (0.188)
SA _i	0.498** (0.199)	0.831*** (0.217)	0.512** (0.211)	0.479** (0.210)	2.045*** (0.312)	3.194*** (0.327)	3.367*** (0.347)	3.294*** (0.348)
Ln(Stage)	2.651*** (0.041)	2.706*** (0.043)	2.489*** (0.041)	2.461*** (0.041)	3.498*** (0.055)	3.710*** (0.059)	3.686*** (0.060)	3.668*** (0.059)
Post	-1.165*** (0.209)				-1.214*** (0.240)			
Treat × Post	0.374 (0.236)	0.355 (0.237)	-0.012 (0.226)	-0.024 (0.225)	0.215 (0.273)	0.253 (0.273)	0.409 (0.277)	0.429 (0.277)
Treat × SA _i × Post	0.911*** (0.227)	0.919*** (0.227)	0.873*** (0.220)	0.887*** (0.219)	0.476 (0.359)	0.372 (0.359)	0.117 (0.376)	0.169 (0.377)
Time FE	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Task FE	No	No	Yes	Yes	No	No	Yes	Yes
Industry FE	No	No	No	Yes	No	No	No	Yes
Observations	113,059	113,059	113,059	113,059	71,823	71,823	71,823	71,823
R ²	0.05	0.05	0.13	0.14	0.07	0.07	0.12	0.13
Within R ²		0.05	0.04	0.04		0.07	0.06	0.06

Clustered (Project ID) standard-errors in parentheses

Signif. Codes: ***: 0.01, **: 0.05, *: 0.1